

ASG Airlines: End-to-End Data Engineering Case Study

Problem Statement

ASG Airlines, a nationwide carrier, operates flights across multiple cities through its booking platforms, scheduling systems, and airport logs. The organization collects operational flight data from these various systems; however, inconsistencies in the dataset such as corrupted flight identifiers, inconsistent time formats, missing values, and unhandled overnight (cross-day) flight scenarios have created challenges in generating accurate operational insights.

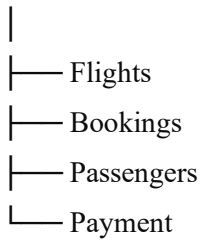
Architecture



Python and Pandas are used for ingestion, cleaning, transformation, validation, and aggregation. The resulting cleaned datasets are used as the source for Power BI

Data Flow:

Excel Workbook



Data Ingestion



Data Quality Checks



Cleaning & Standardization



Transformation



Validation



PII-Safe Datasets



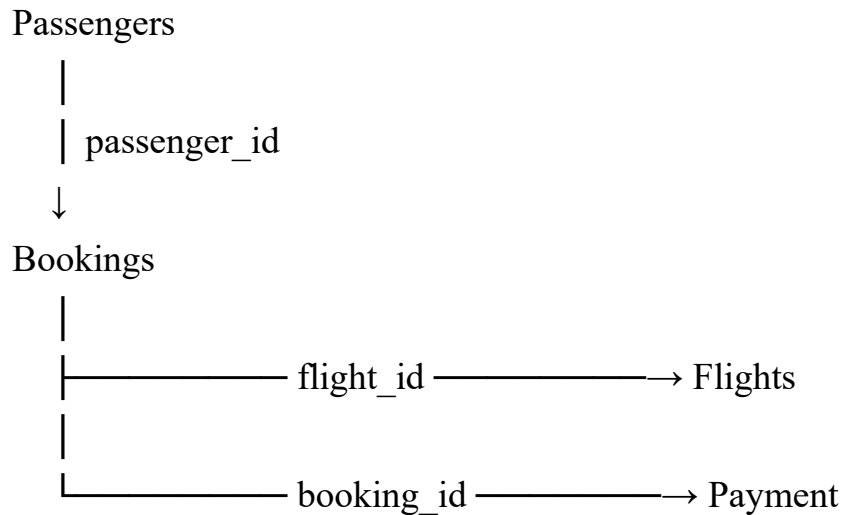
KPI Generation



Power BI

The four datasets are related using their identifiers. Flights are connected to bookings through `flight_id`, passengers are connected to bookings through `passenger_id`, and payments are connected to bookings through `booking_id`.

Realationships:



Data cleaning Strategy

The following cleaning operations were performed:

Duplicate Records

Complete duplicate rows in the flights dataset were removed.

Missing Categorical Values

Missing values in selected categorical/text fields were replaced with UNKNOWN so that the records were retained without creating unsupported values.

Missing Payment Amounts

Missing payment amounts were retained as missing values instead of replacing them with zero because a missing payment amount does not necessarily mean that no payment was made.

Data Type Standardization

Date and time columns were converted to appropriate datetime formats, and payment amounts were converted to numeric values.